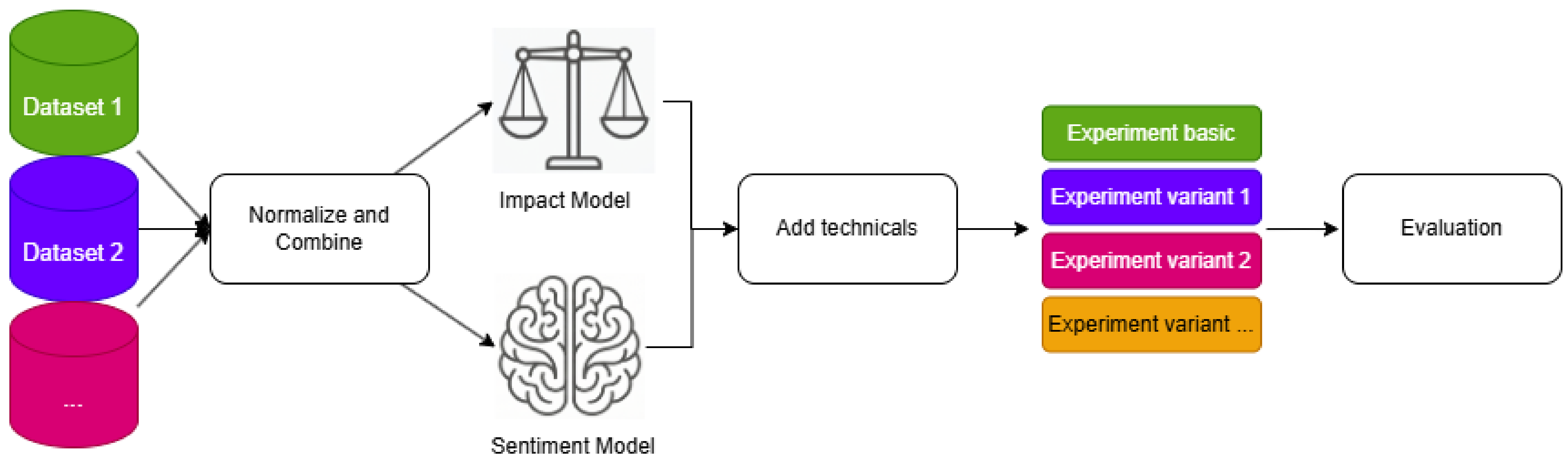


Introduction:

- Rapid advances in Natural Language Processing (NLP) [1] have created new opportunities for understanding the stock market movements.
- Efficient Market Hypothesis (EMH) states that all information is priced in [2]
- Mixed results in literature [3] [4] [5]
- Research Question: Can daily aggregated sentiment extracted from news headlines improve prediction accuracy for S&P 500 stock prices?

Methods:

- Combine multiple headline datasets
- Use LLMs (e.g. FinBERT) to extract investor mood
- Perform denoising via an impact score model (Llama 3.1 Instruct (8B))
- Use this feature in classification and regression tasks
- Cross validate on multiple evaluation models and experiment variations



Results:

(performance compared to predictions using only yesterday's price changes)

Model	Δ RMSE	Δ MAE	ΔR^2	
ARIMA	+0.000010	+0.000015	-0.003668	
LSTM	0.000000	0.000000	-0.000741	
GRU	-0.000004	0.000000	+0.000308	
MLR	+0.000021	+0.000012	-0.003257	
Mean-Baseline	0.000000	0.000000	0.000000	
Model	Δ Accuracy	Δ Precision	Δ Recall	Δ F1-Score
XGBoost	-0.015416	-0.010901	-0.033019	-0.020982
MLogR	+0.028750	+0.180575	+0.270727	+0.210957
Always Up	0.000000	0.000000	0.000000	0.000000

Figure 1: Performance delta with sentiment as the sole predictor

Model	Δ RMSE	Δ MAE	ΔR^2	  
ARIMA	+0.000007	-0.000005	-0.001799	
LSTM	-0.000002	-0.000005	-0.000165	
GRU	-0.000011	-0.000012	+0.001696	
MLR	+0.000017	-0.000001	-0.000863	
Mean-Baseline	0.000000	0.000000	0.000000	
Model	Δ Accuracy	Δ Precision	Δ Recall	Δ F1-Score
XGBoost	-0.003333	-0.001110	-0.027280	-0.012707
MLogR	+0.053334	+0.226538	+0.481488	+0.335258
Always Up	0.000000	0.000000	0.000000	0.000000

Figure 2: Performance delta with denoised sentiment as the sole predictor

In a fair, non-lookahead setup, there is no consistent benefit in using the engineered sentiment feature, suggesting that the EMH holds for the performed experiments.

References:

- [1] Araci (2019) "Finbert: Financial sentiment analysis..."; [2] Malkiel & Fama (1970) "EFFICIENT CAPITAL MARKETS..."; [3] Davidovic & McCleary (2025) "News Sentiment and Stock Market Dynamics..."; [4] Zhang et al. (2018) "A novel data-driven stock price..."; [5] Farimani et al. (2022) "Investigating the informativeness..."